# Living Breakwaters in High-Energy Environments: Modeling Circulation and Transport around a Remnant Submerged Breakwater

Alejandro Angel[1*] Bernabe Gomez[1,2] Timu Gallien[1]

[1]Department of Civil and Environmental Engineering, University of California, Los Angeles - [2]University of Gibraltar | *alejandroangel@ucla.edu

## Introduction

Breakwaters protect billions of dollars in coastal infrastructure and ecosystem services worldwide. Most existing breakwaters were constructed in the early–mid 20th century and are operating beyond their intended service life. Intensifying storms and rising sea levels may damage existing breakwaters. In 1983 energetic waves toppled Santa Monica's emerged breakwater, resulting in a remnant submerged breakwater. Submerged rubble mound breakwaters modify surfzone circulation which cannot be resolved using conventional phase-averaged models. Downdrift erosion, lee-side stagnation, Infragravity (IG) resonance, and habitat fragmentation remain poorly characterized in **high-energy, swell-dominated environments**.

In the United States, NOAA catalogs over 200 living shoreline projects, nearly all are located in estuarine or riverine environments and subject to relatively minimal wave energy (NOAA, 2026). Few living shoreline structures have been deployed in energetic wave environments. Hybrid "living" breakwater retrofits present an attractive adaptation pathway mitigating coastal hazards (e.g., erosion, flooding) and providing enhanced intertidal habitats. However, there is a paucity of nature based breakwater design guidance. The Santa Monica remnant submerged breakwater provides an exceptional case study for living breakwater implementation.

Here, a coupled **SWAN–FUNWAVE-TVD–Delft3D** framework is used to characterize tidal and wave driven circulation. High-resolution hydrodynamic model outputs drive a Total Variation Diminishing (TVD) passive-tracer advection-diffusion model, to quantify how IG motions and wave-shadow trapping at the breakwater influence near-field residence time and pollutant fate under quiescent and storm conditions.

## Study Area and Observations

The Santa Monica Breakwater lies 600 m offshore of Santa Monica Pier (California). Two moorings, each carrying a pressure sensor and an Acoustic Doppler Current Profiler ADCP, were deployed November 2024-January 2025 at locations: **Lee** (sheltered behind the structure) and **Side** (open-coast control). A major swell event peaked on 23 Dec 2024 ($H_s > 1.5$ m); calm conditions prevailed 21 Dec 2024.

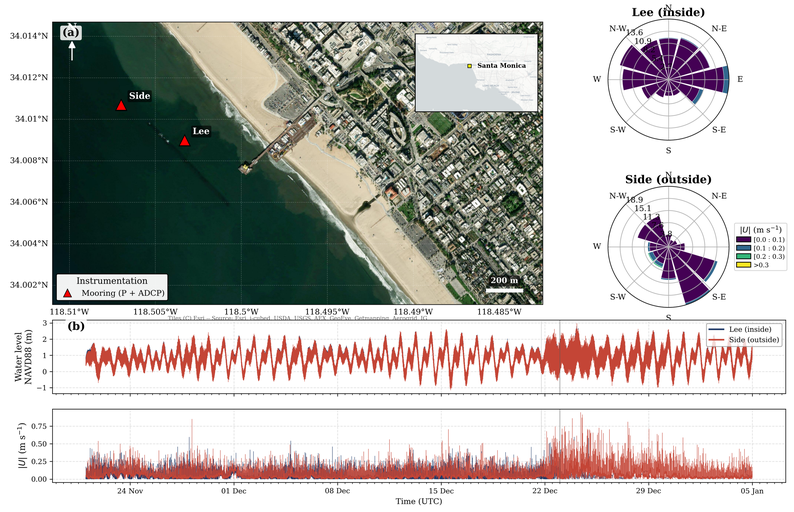


**Figure 1.** (a) Satellite view of the breakwater with the two moorings (Side and Lee). (b) Water level (NAVD88) and depth-averaged current magnitude $|U|$ (m/s) at both moorings. Storm event peaks 23 Dec 2024. Current roses (right) show the directional asymmetry between Lee (predominantly N–S) and Side (predominantly S-E).

## Methodology

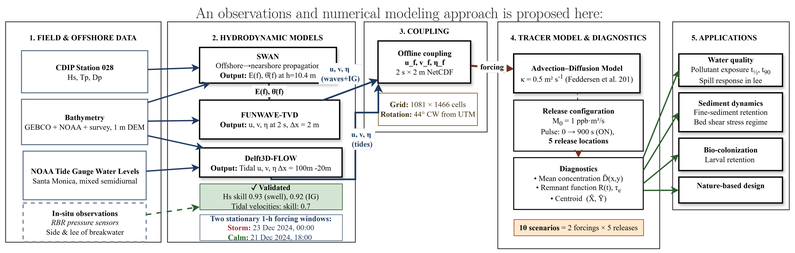


**Figure 2.** End-to-end workflow. Field and offshore data force the coupled SWAN–FUNWAVE-TVD–Delft3D framework, whose composite output drives a TVD advection–diffusion tracer model.

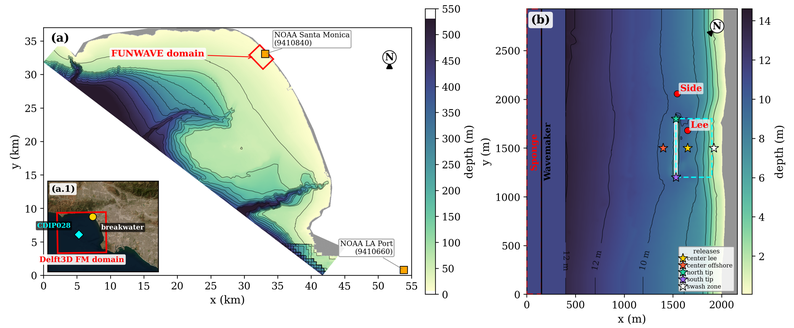


**Figure 3.** (a) Delft3D and SWAN domain over Santa Monica Bay; inset shows offshore CDIP-028 spectral boundary station. (b) FUNWAVE-TVD subdomain with wavemaker (black), sponge (red), observation stations (red), five tracer release locations (stars), and lee control region (cyan) to study residence time.

A coupled hydrodynamic framework is proposed to capture high-resolution circulation around the breakwater:

**Delft3D** solves tidal circulation in a flexible mesh grid over Santa Monica Bay with 100m to 20m resolution and NOAA-water levels as boundary conditions. Results were validated against observations, reaching skill scores of 0.99 for water levels and 0.6-0.8 for tidal velocity components.

**SWAN** (Simulating Waves Nearshore) solves the propagation of wave climate into nearshore using a 20m resolution grid covering a significant part of the bay, under a JONSWAP bottom friction formulation. Offshore wave forcing comes from spectral measurements of Significant Wave Height Hs and Peak Period Tp of the CDIP 028 Station located offshore of Santa Monica bay. Simulated Hs and Tp are in good agreement with observations at the side of the breakwater as seen in figure 4.

**FUNWAVE-TVD** [1] resolves nonlinear wave processes within the nearshore and breakwater using a 2m resolution grid with 1081 x 1466 cells with an internal wavemaker forcing[2]. FUNWAVE results were validated against water level observations using independent 1h simulation windows, reaching skill scores of 0.8-0.9 for Hs on both swell and IG frequency bands.

Composite forcing $(\eta, u, v)$ is stored at **2 s and 2 m resolution** on a FUNWAVE grid rotated 44° from UTM east.

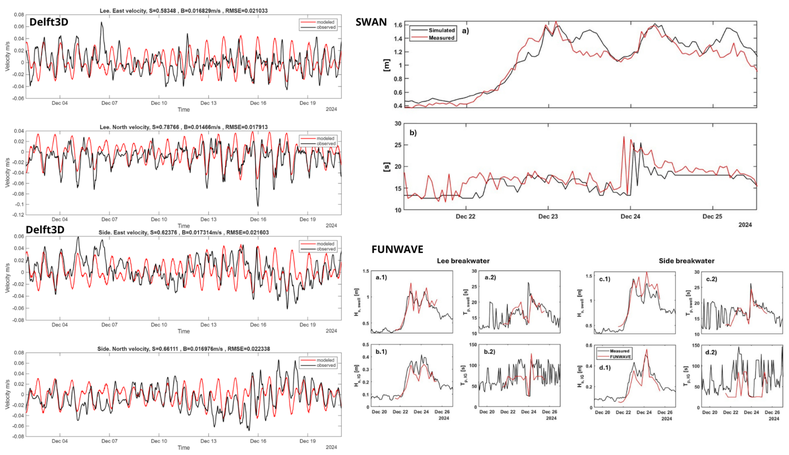


**Figure 4.** Model–observation skill. **Delft3D** tidal velocities at Lee/Side: $S = 0.58$–$0.79$, RMSE $\leq 2.2$ cm s$^{-1}$. **SWAN** $H_s$ and $T_p$ track the storm peak. **FUNWAVE-TVD** $H_{s,\text{swell}}$ and $H_{s,\text{IG}}$ at both stations: skill $\approx 0.93$ (swell), $\approx 0.92$ (IG).

**Tracer transport.** To investigate transport mechanisms, we developed a Depth-integrated conservative advection-diffusion following Clark et al. [3]:

$$\frac{\partial (h+\eta)C]}{\partial t} + \nabla \cdot [(h+\eta)\mathbf{u}C] = \nabla \cdot [(h+\eta)K\nabla C] + S_q$$

Where C is the depth-averaged tracer concentration, h is the still water depth, $\eta$ is the free surface elevation, U is the depth-averaged velocity for each direction, K is the horizontal eddy diffusivity and Sq is the conservative source term.

We discretize the equation using a TVD finite-volume scheme (Superbee flux limiter) with forward-Euler time stepping at $\Delta t = 0.15$ s and constant horizontal diffusivity $K = 0.5\,\text{m}^2\text{s}^{-1}$ [4]. The tracer source is a normalized point flux of $M_0 = 1\,\text{ppb}\cdot\text{m}^3/\text{s}$ applied for the first 900 s of each 1-hour simulation. For transport studies we tested 2 forcing windows (Table 1) and five release locations around the breakwater (Figure 3) for a total of 10 simulations, to assess different configurations.

| Forcing window | UTC |
|---|---|
| Storm peak (high $H_s$, energetic IG) | 23 Dec 2024, 00:00–01:00 |
| Calm (low $H_s$, tidal-dominated) | 21 Dec 2024, 18:00–19:00 |

**Table 1.** Two 1-h stationary forcing windows.

## Results

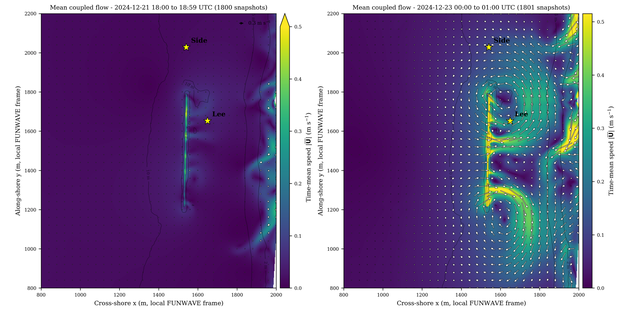


**Figure 5.** Time-mean coupled flow $|\overline{\mathbf{U}}|$ (color) and direction (arrows). **Calm** (left): essentially quiescent, weak tidal residual. **Storm** (right): coherent counter-rotating cells anchored at the breakwater tips, with peak return-flow around 0.3–0.5 m s$^{-1}$.

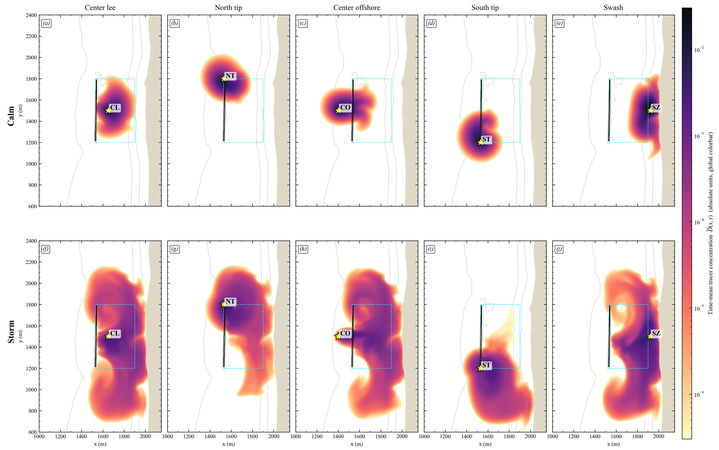


**Figure 6.** Time-averaged tracer concentration $\overline{D}(x, y)$ over $t \in [900, 3600]$ s. **Calm** (top, a–e): plumes remain near release sites, with footprint $<300$ m. **Storm** (bottom, f–j): plumes spread $\sim$1 km after 1 hour

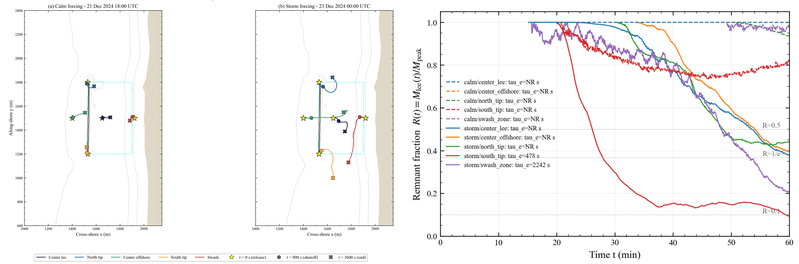


**Figure 7–8. (Left)** Plume centroid trajectories. Calm centroids drift less than 100 m by $t = 3600$ s. Under storm forcing, centroids exhibit *return loops* — the kinematic fingerprint of the IG-anchored cells in Figure 5. **(Right)** Remnant fraction $R(t) = M_{\text{lec}}(t)/M_{\text{peak}}$ in the lee control region. Under calm forcing, $R(t) \geq 0.8$ for all releases. Under storm forcing, south-tip and swash-zone releases flush ($\tau_e = 478$ s and 2242 s respectively); center-lee and north-tip releases remain trapped.

## Conclusions

- In-situ high resolution hydrodynamic observations were used to validate a nested near-field breakwater circulation model
- The coupled **SWAN–FUNWAVE-TVD–Delft3D** framework accurately reproduces water level (skill=0.99), tidal (skill=0.6-0.8) and wave driven currents (skill=0.9).
- Wave driven circulation dominates near field velocities
- Under energetic wave conditions, two strong circulation cells form at the ends of the breakwater.
- This circulation pattern effectively **partitions the lee in two compartments**. These circulation cells serve to trap and/or eject tracer, depending on wave direction and tidal phase.
- Under calm wave conditions, the lee is hydrodynamically quiescent (i.e., velocities $\leq 0.1$ m/s)
- The remnant breakwater has low elevation passages between displaced stone which form jets resulting in offshore tracer releases propagating shoreward and becoming trapped in the lee (Figure 5).
- **Design implication:** Although the breakwater mitigates wave energy, it may encourage pollutant trapping.

## Data & Animations!

Code, data & animations:
github.com/
al3jandroangel
alejandroangel@ucla.edu

## Limitations & Future Work

- Tracer skill inherits from hydrodynamic validation; direct tracer comparison reserved for follow-on study.
- Depth-integrated formulation; justified by well-mixed CTD casts, buoyancy effects neglected.
- 1-h stationary forcing; longer-window runs with tidal superposition in progress.
- Parametric sweep over candidate hybrid retrofit geometries is the next step.

## Acknowledgements

This study was funded by The Bay Foundation (Award 24-08). We would like to gratefully acknowledge Gabriela Carr, Davis Kim, Michael Angelis, Quentin Laporte, Margit Maple, the City of Santa Monica and The Bay Foundation crew who contributed and participated in the field work that made this work possible. We thank Kees Netherhof and Delphine Hypolite for their valuable support on tidal modeling.

## Key References

[1] Shi et al. *A high-order adaptive time-stepping TVD solver for Boussinesq modeling.* Ocean Modelling 43, 2012.
[2] Wei, Kirby & Sinha. *Generation of waves in Boussinesq models using a source function method.* Coastal Eng. 36, 1999.
[3] Clark, Feddersen & Guza. *Modeling surf zone tracer plumes II: Transport and dispersion.* JGR Oceans 116, 2011.
[4] Feddersen et al. *Observations and modeling of a tidal inlet dye tracer plume.* JGR Oceans 121, 2016.
[5] MacMahan, Thornton & Reniers. *Rip current review.* Coastal Eng. 53, 2006.